

DSA210 Final Report — Digital Escapism & Socio-Economic Stress Analysis

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Digital Escapism & Socio-Economic Stress Analysis

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This report is the final, consolidated deliverable for the DSA210 term project. It bundles the work of the three earlier milestones (proposal, EDA + hypothesis tests, ML methods) into a single document. All code, data, and figures referenced here live in the same GitHub repository.

1. Motivation

Turkey has experienced sustained macroeconomic volatility between 2023 and 2025: rapid currency depreciation, double-digit inflation, and large swings in the Borsa Istanbul 100 index. Behavioural-economics and media-studies literature suggests that, when households face financial stress, they often substitute cheap or already-paid-for entertainment for more expensive leisure. PC gaming on Steam — where most titles are owned permanently and play time is essentially free at the margin — is a natural candidate for that substitution.

This project investigates the **digital escapism hypothesis** in Turkey:

When the Turkish economy deteriorates, does aggregate engagement with Steam-based gaming in Turkey increase?

Answering this question requires linking two data ecosystems that are not normally combined: financial-market data (USD/TRY, BIST100) and gaming activity (Steam / Google Trends). Even a partial answer is useful: it informs how digital-entertainment demand responds to macroeconomic shocks in an emerging market.

2. Data Sources & Collection

#	Source	Variable	Coverage	Collection tool
1	SteamDB (steamdb.info)	CS2 (app 730) daily concurrent players, global	2023-01 → 2025-12	Manual CSV download
2	Yahoo Finance via <code>yfinance</code>	USD/TRY exchange rate, daily close	2023-01 → 2025-12	<code>yfinance</code> Python package

#	Source	Variable	Coverage	Collection tool
3	Yahoo Finance via <code>yfinance</code>	BIST100 index, daily close	2023-01 → 2025-12	<code>yfinance</code> Python package
4	Google Trends via <code>pytrends</code>	TR search interest for <code>steam</code> , <code>cs2</code> , <code>counter strike</code>	2023-01 → 2025-12	<code>pytrends (geo='TR')</code>
5	Google Trends via <code>pytrends</code>	TR search interest for <code>ekonomik</code> <code>kriz</code> , <code>enflasyon</code> , <code>dolar kur</code> , <code>issizlik</code>	2023-01 → 2025-12	<code>pytrends (geo='TR')</code>

Enrichment rationale. The original idea was to study Steam activity against a single financial series. To meet the guidelines’ enrichment requirement, I added (a) the BIST100 equity index, (b) a composite Google Trends “economic-stress” index built from four Turkish-language stress keywords, and (c) Turkey-specific Google Trends interest in Steam/CS as a country-level gaming proxy. Together these turn a 2-series dataset into a 5-series, multi-source dataset suitable for both hypothesis testing and multivariate regression.

Proxy disclosure. Steam does not publish country-level concurrent player counts. The Turkey-specific gaming signal in this project is Google Trends search interest in Turkey for Steam-related queries, which is the best openly available proxy. Global CS2 concurrent players from SteamDB is kept as a *global reference* series.

Resulting dataset. Daily series are resampled to monthly frequency, producing **36 monthly observations** (2023-01 → 2025-12) with no missing values after cleaning. See `Final/data/merged_dataset.csv` .

3. Data Pipeline

```
data_collection.py → raw CSVs (usdtry, bist100, google_trends_TR,
steam_cs2_clean)
data_processing.py → merged_dataset.csv (monthly, z-scored, crisis flag)
EDA notebook      → hypothesis tests + EDA figures
ML notebook       → ml_features.csv + holdout performance + ML figures
FINAL_REPORT.md   → this document (consolidates everything)
```

Key transformations in `data_processing.py` :

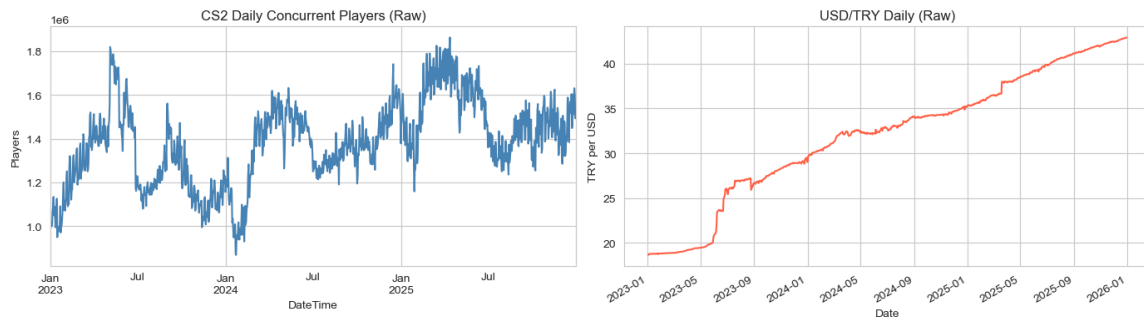
- 1. **Resample** every series to monthly mean.
- 2. **Inner join** on month index → 36 aligned rows.
- 3. **Z-score normalisation** (`*_z` columns) for cross-series comparability.
- 4. **Crisis flag** (`crisis_period = 1` if monthly USD/TRY ≥ 75th percentile, else 0).
- 5. **Composite stress index** = mean of the four z-scored Turkish stress keywords.

The pipeline is fully deterministic (no random sampling, no model randomness at this stage) and reproducible with two `python` commands (see §10).

4. Exploratory Data Analysis

Full notebook: `34314_AliEfe0kudan_EDA.ipynb` (root of repo). Figures referenced here live in `Final/figures/`.

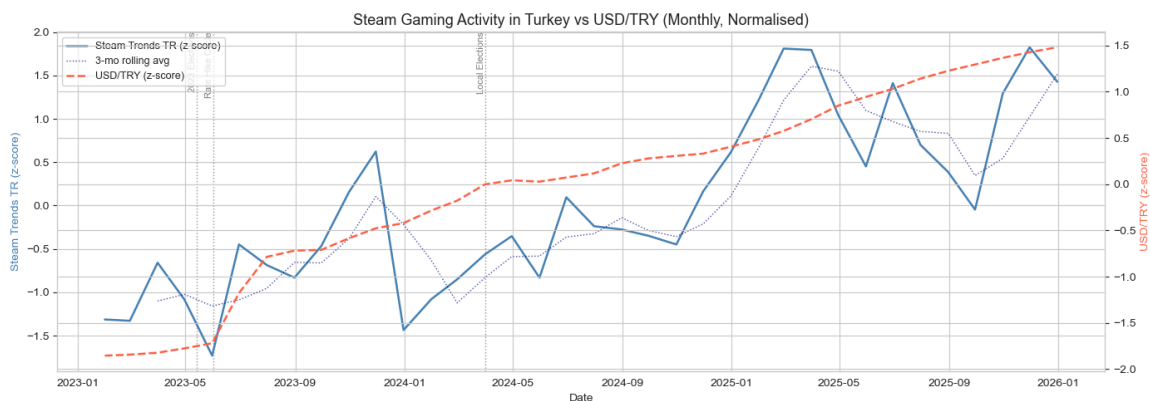
4.1 Raw series



Raw series

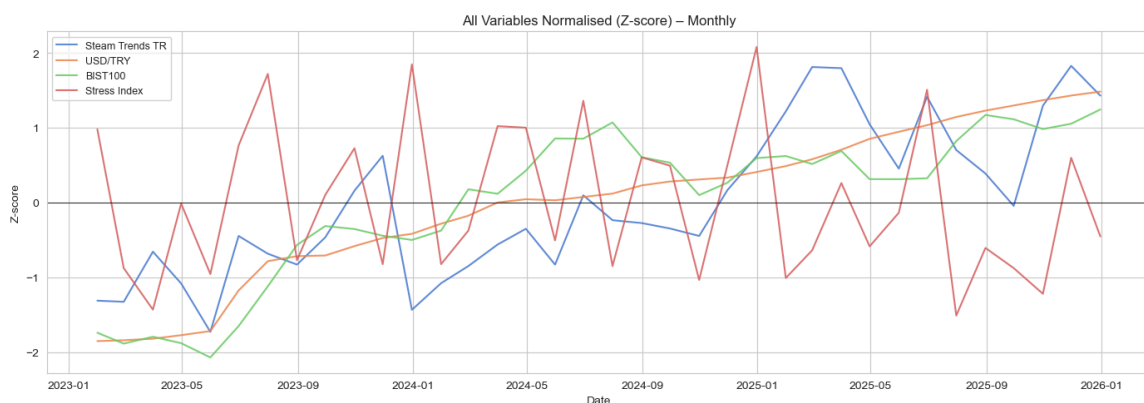
USD/TRY moves almost monotonically upward over the window; BIST100 is trend-up with two distinct corrections; both gaming series (global CS2, TR Steam Trends) are visually correlated with USD/TRY's trajectory.

The dual-axis overlay below makes the co-movement between USD/TRY and the TR Steam-Trends proxy explicit on the original scales (left axis: TRY per USD, right axis: Trends search interest, 0–100):



USD/TRY vs TR Steam Trends — dual axis

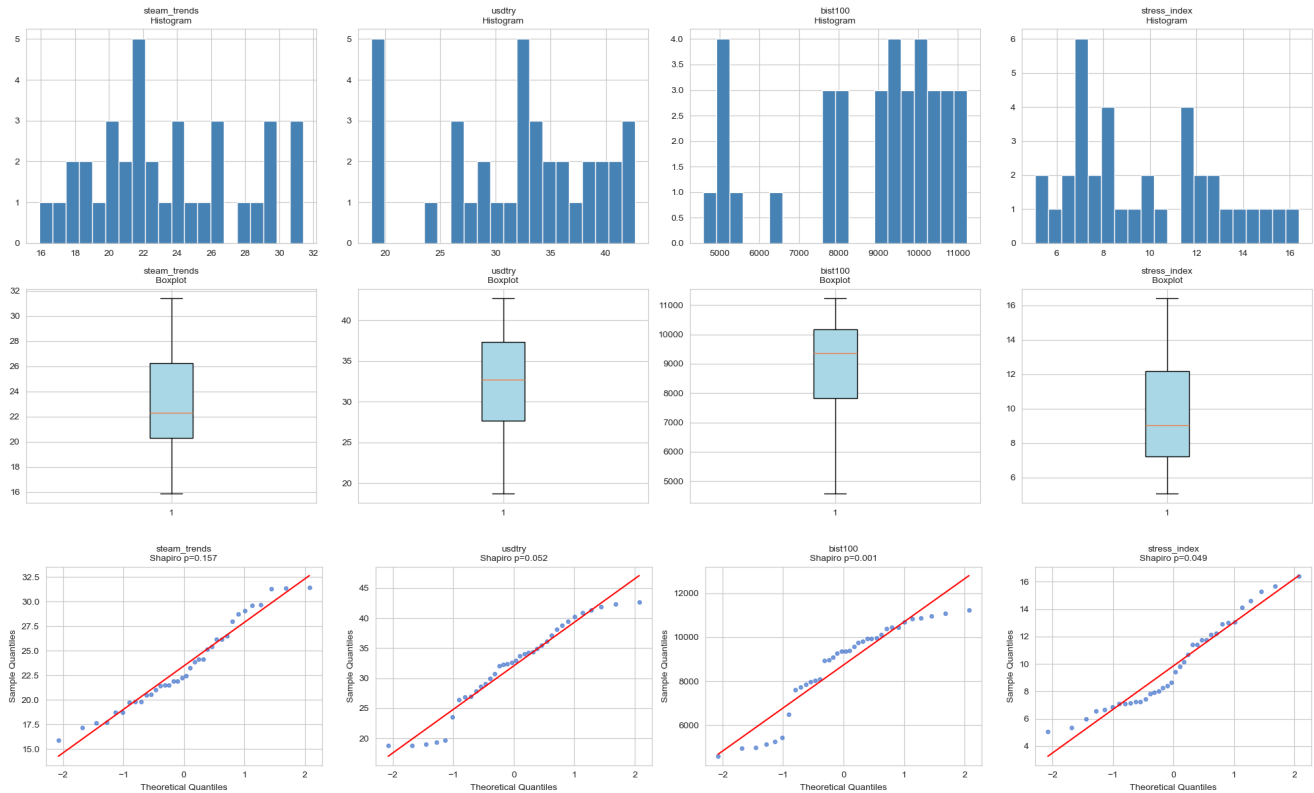
4.2 Z-scored view



All series z-scored

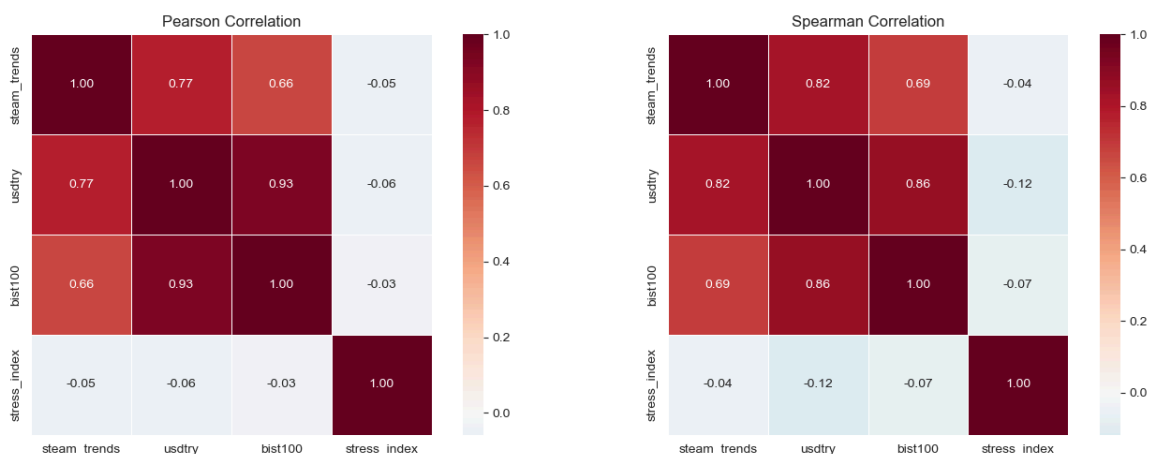
Normalising to z-units lets the five series share an axis. USD/TRY and the TR Steam-Trends proxy move together; the global CS2 player count is less synchronised, consistent with it being a worldwide aggregate rather than a Turkey-specific signal.

4.3 Distributions and normality



Most variables are mildly non-normal — important justification for using Mann-Whitney U (rank-based, no normality assumption) for H2 in §5.

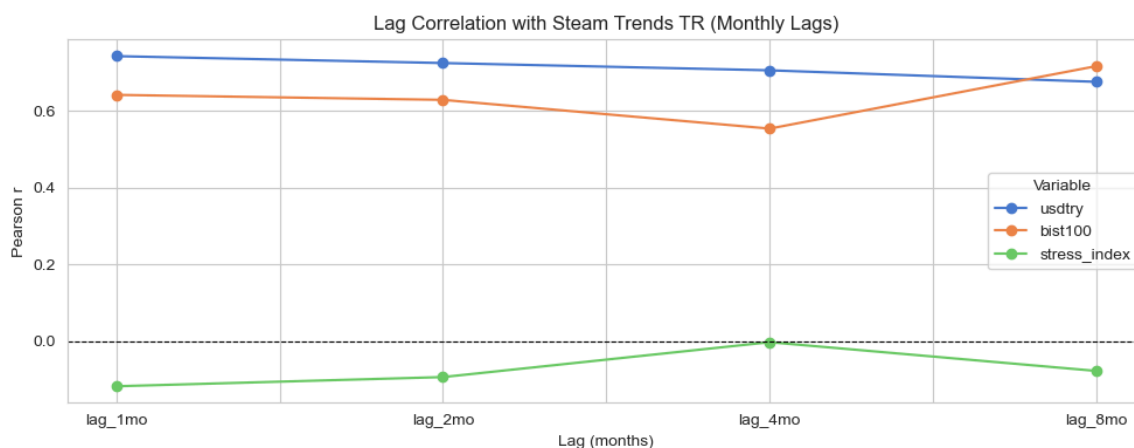
4.4 Correlation structure



Correlation heatmap

USD/TRY ↔ TR Steam Trends shows the strongest cross-series correlation in the matrix. BIST100 correlates with USD/TRY (both reflect the same underlying macro regime). The composite stress index, surprisingly, does **not** load on the gaming proxy — foreshadowing the H3 result.

4.5 Lag structure



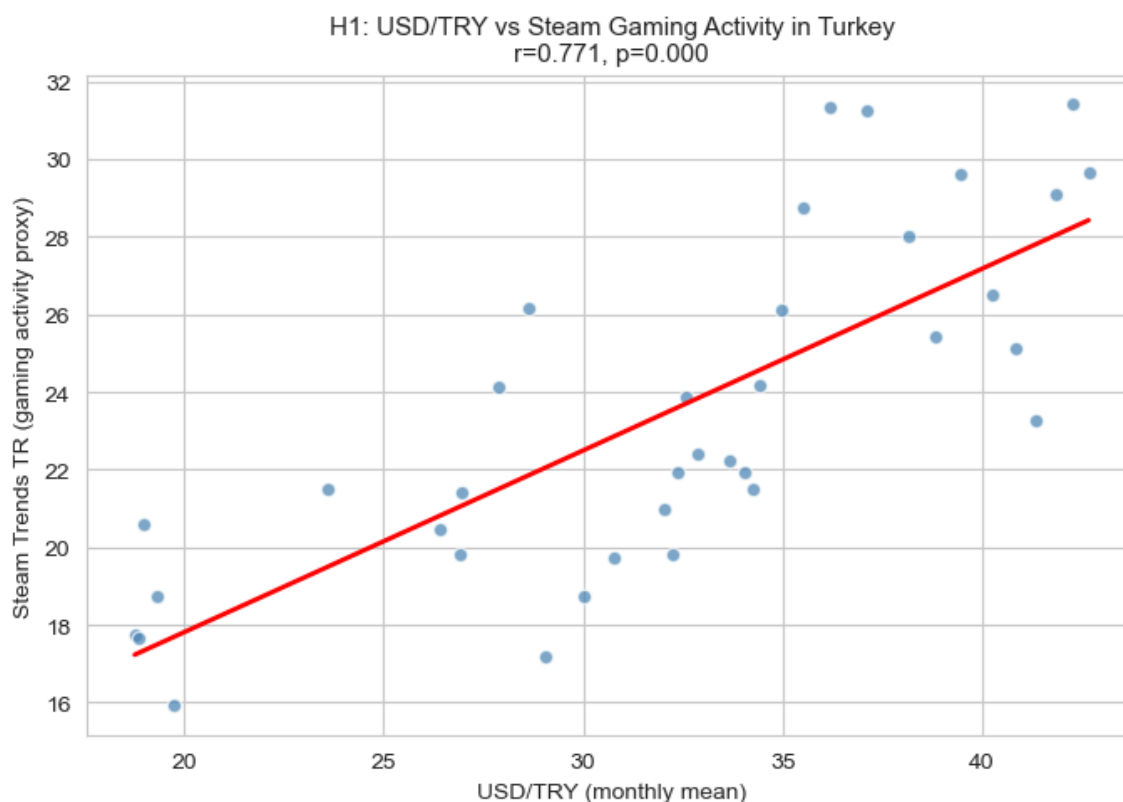
Lag correlation

Cross-correlation between `usdtry_z` and `steam_trends_z` peaks at lag 0 and remains positive at lags 1–2. This motivates including 1- and 2-month USD/TRY lags as ML features (§6).

5. Hypothesis Tests

Three preregistered hypotheses, tested on the monthly merged dataset.

H1 — USD/TRY ↔ TR Steam gaming activity (positive correlation)

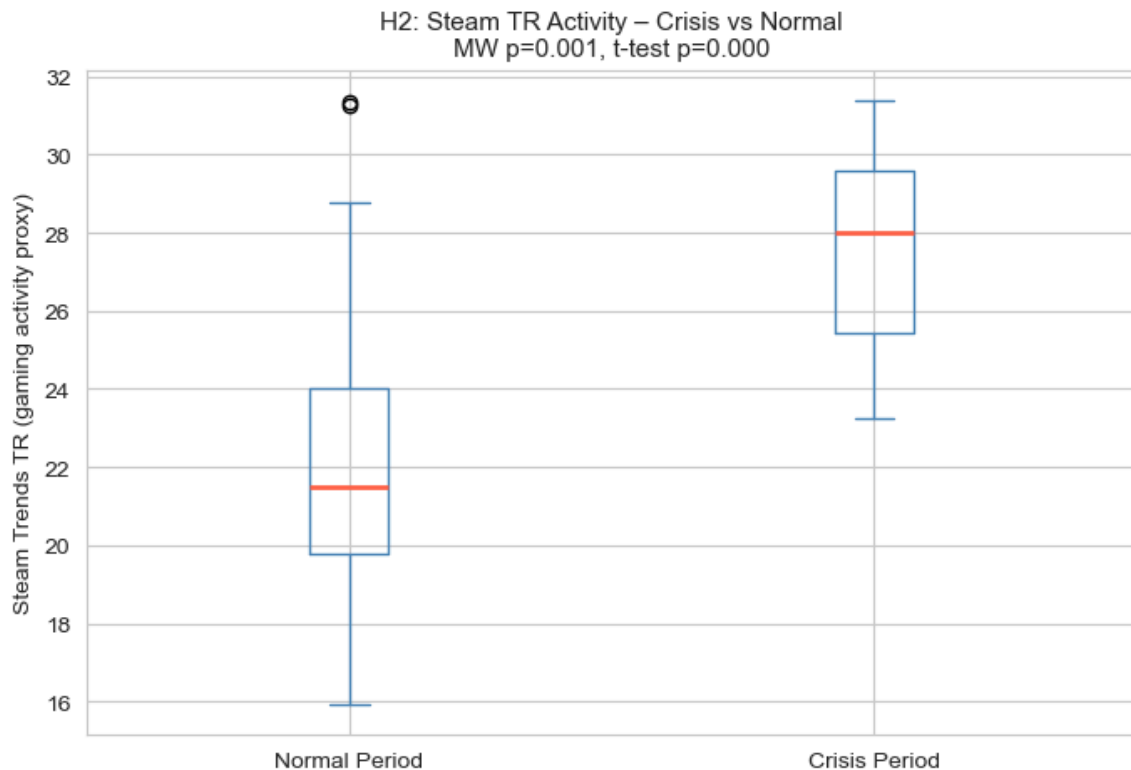


H1 scatter

- Test: Pearson correlation, two-sided.

- Result: $r = 0.771$, $p < 0.001$.
- Conclusion: **Supported**. As USD/TRY rises, Turkey-specific Steam search interest rises with a strong, statistically significant linear relationship.

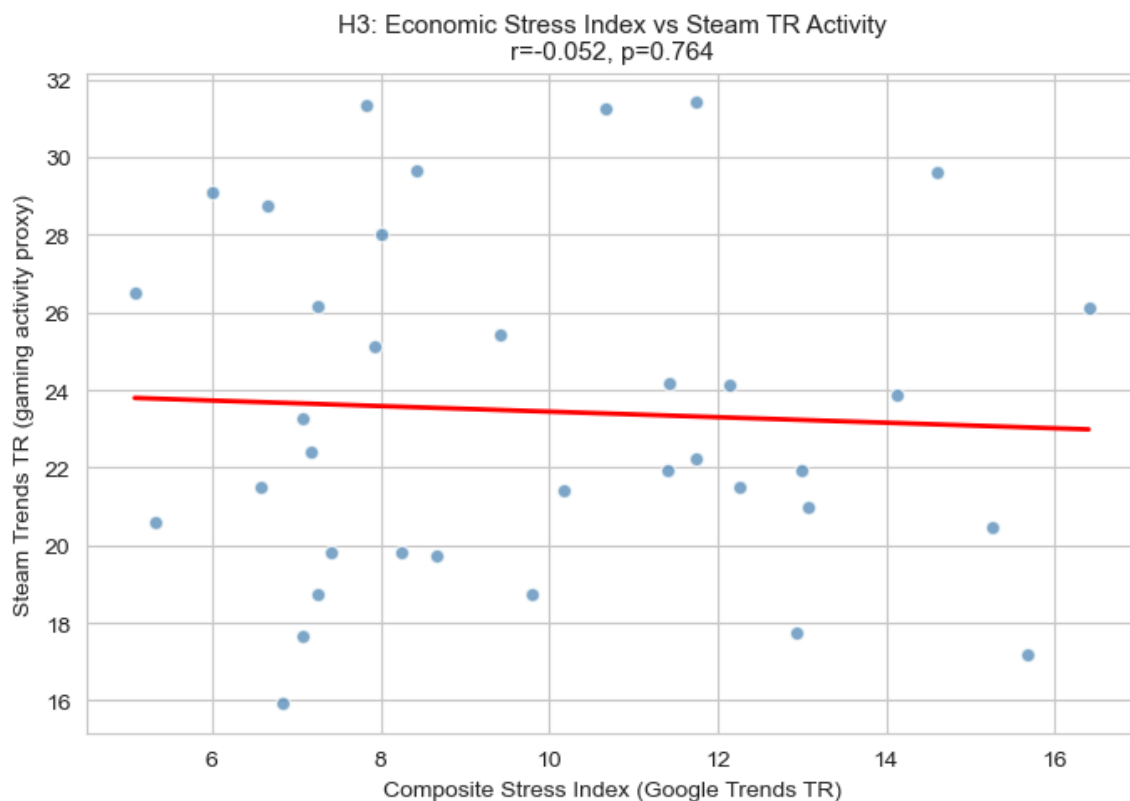
H2 — Crisis vs. normal periods (gaming higher in crisis)



H2 boxplot

- Definition of crisis month: USD/TRY $\geq$ 75th percentile of the 36-month series (9 crisis months, 27 normal months).
- Test: Mann-Whitney U (two-sided), as run in the EDA notebook.
- Result: $U = 213.0$, $p = 0.0009$. (One-sided p in the *crisis* > *normal* direction is $p = 0.0004$; reported value is the two-sided one to match the notebook output.)
- Conclusion: **Supported**. TR Steam search interest is significantly higher during crisis months than during normal months.

H3 — Composite stress index ↔ TR Steam gaming activity



H3 scatter

- Test: Pearson correlation, two-sided.
- Result: $r = -0.052$, $p = 0.764$.
- Conclusion: **Not supported**. The search-volume-based stress index does not correlate with the gaming proxy. The likely interpretation: people searching for `ekonomik kriz` and people increasing gaming time are different sub-populations responding to the same macro shock; the market price (USD/TRY) is a cleaner stress signal than search-derived sentiment.

Summary

Hypothesis	Test	Statistic	p-value	Verdict
H1	Pearson	$r = 0.771$	< 0.001	Supported
H2	Mann-Whitney U (two-sided)	$U = 213.0$	0.0009	Supported
H3	Pearson	$r = -0.052$	0.764	Not supported

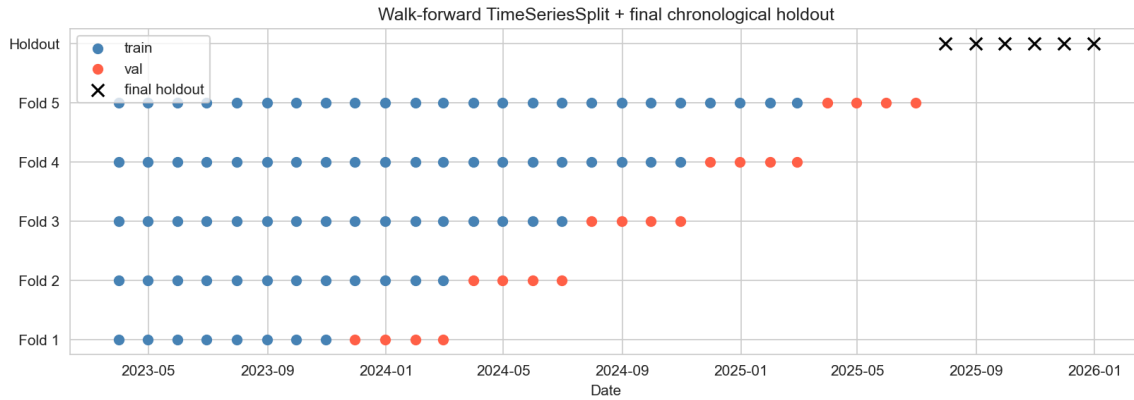
6. Machine Learning Stage

Full notebook: `ML/34314_AliEfe0kudan_ML.ipynb` . Static HTML copy (no need to re-execute):
`Final/34314_AliEfe0kudan_ML.html` . Simpler companion notebook:
`ML/ML_Project/ml_implementation.ipynb` .

6.1 Setup

- **Target:** `steam_trends_z` (z-scored Google Trends TR for steam / cs2 / counter strike).

- **Features (10):** `usdtry_z`, `bist100_z`, `stress_index_z`, `steam_players_z`, `usdtry_lag1`, `usdtry_lag2`, `usdtry_roll3`, `time_index`, `crisis_period`, `usdtry × crisis` interaction.
- **Train/test:** chronological split — **28-month train**, **6-month holdout (2025-07 → 2025-12)**.
- **Validation:** 5-fold `TimeSeriesSplit` on the training set (walk-forward). All scaling fit inside an `sklearn.Pipeline` per fold to avoid leakage.

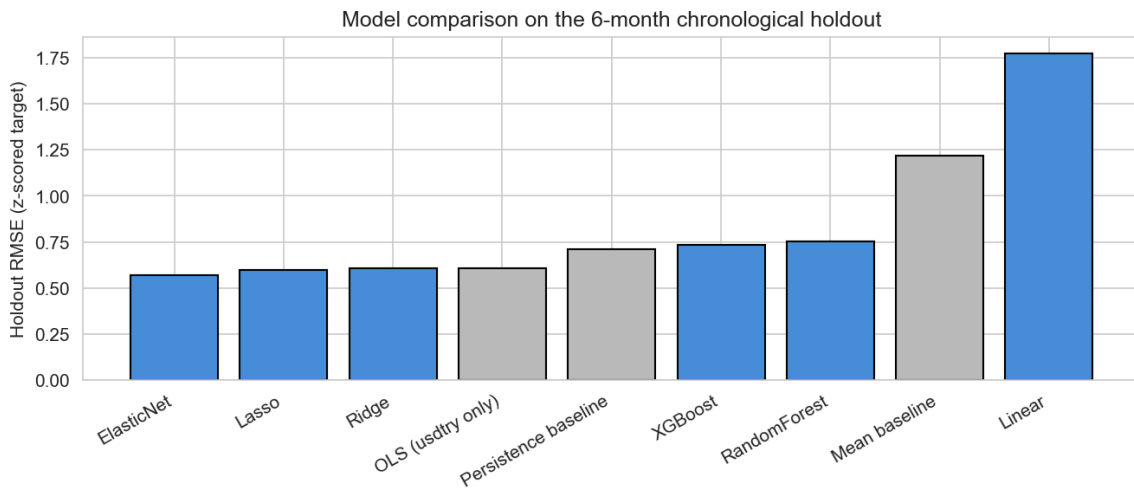


CV split

6.2 Models compared

- Baselines: mean, naïve persistence ($\hat{y}_t = y_{t-1}$), univariate OLS on `usdtry_z`.
- Learned models: Linear Regression, Ridge, Lasso, ElasticNet (α grid), Random Forest, XGBoost (depth grid).

6.3 Holdout performance

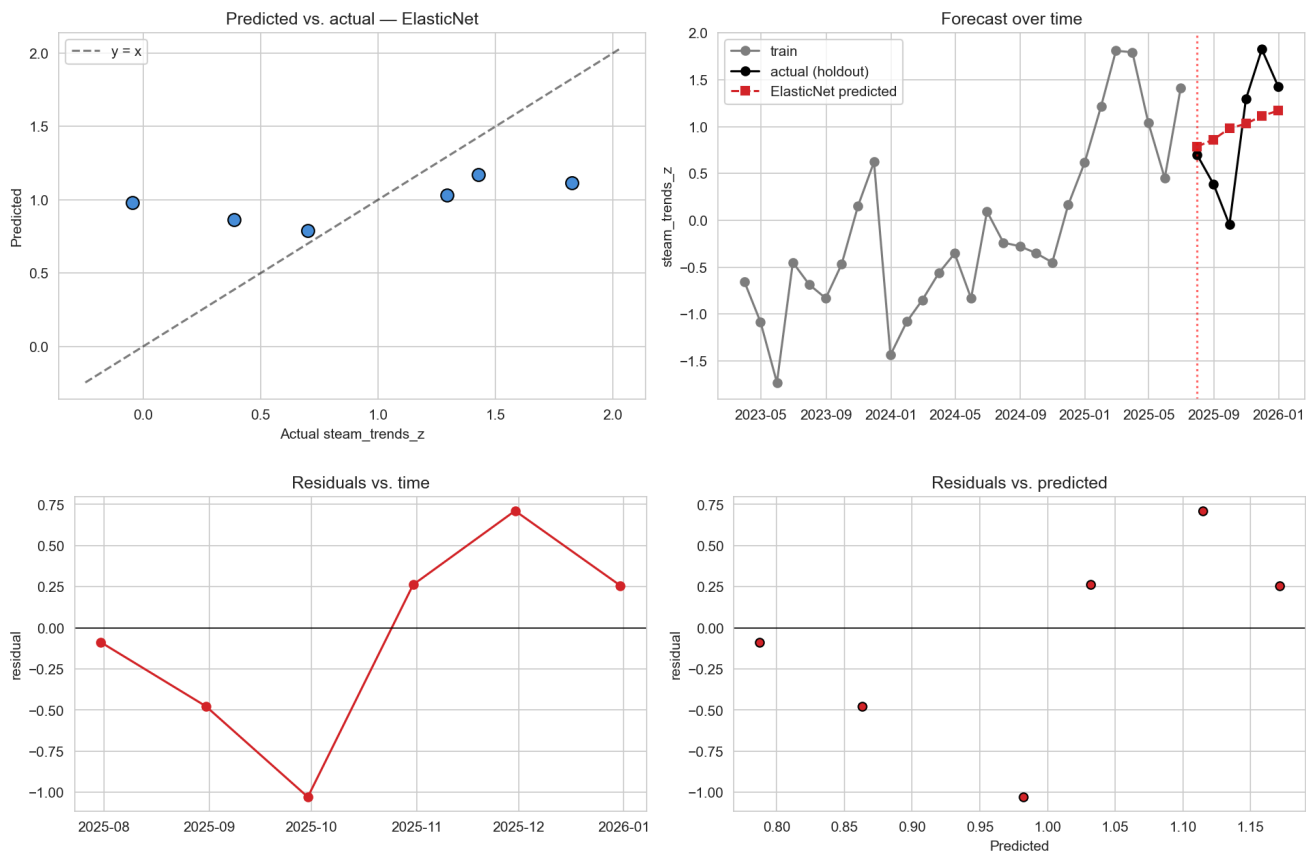


Model comparison

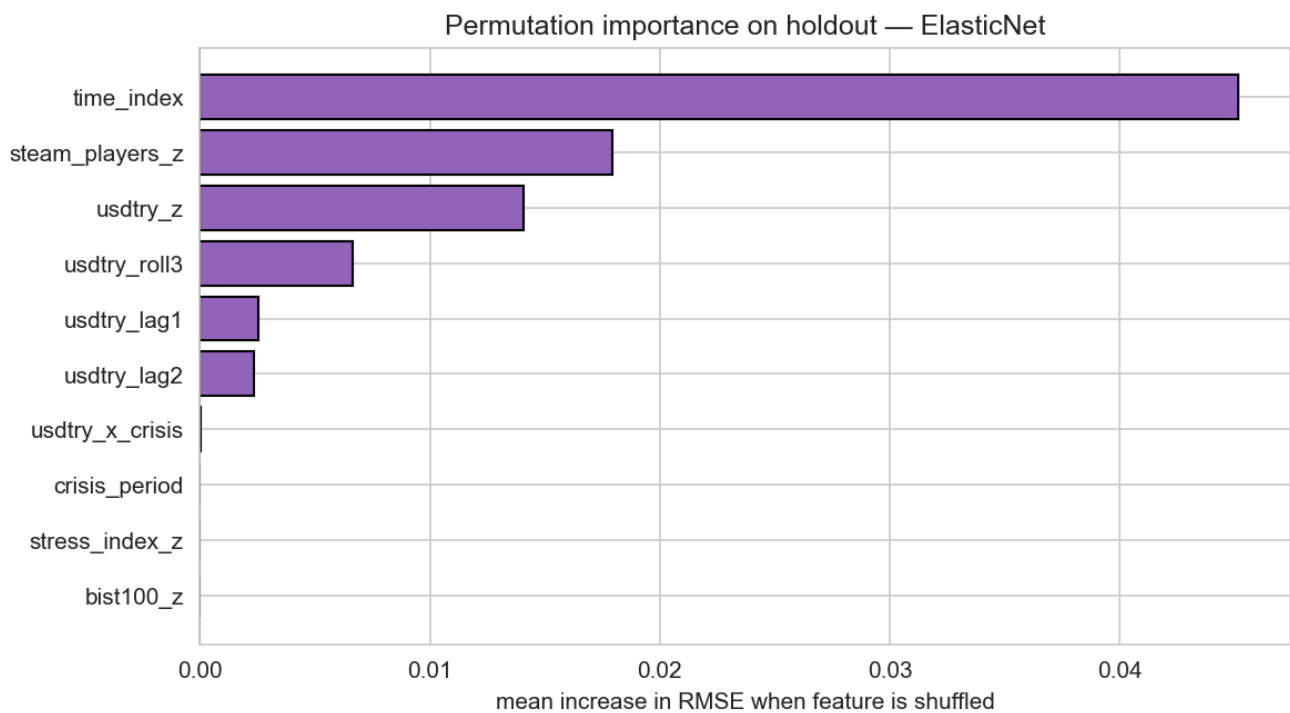
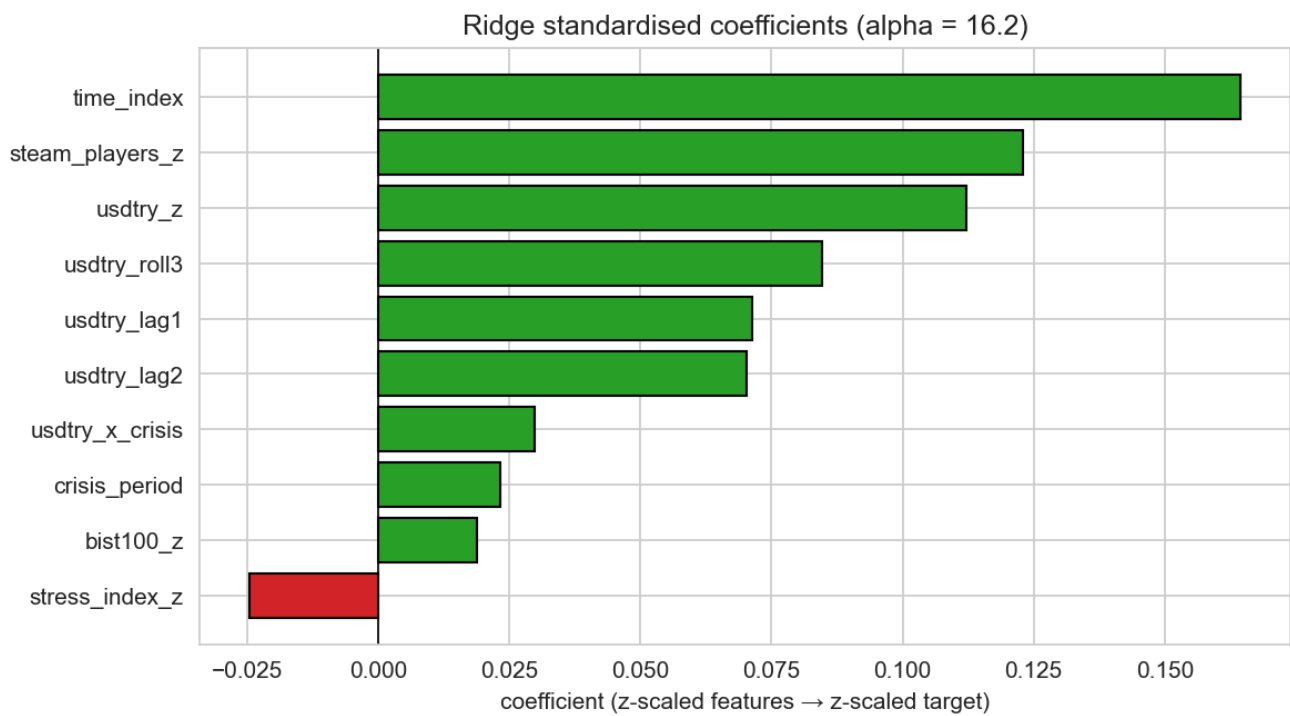
Model	RMSE (z)	MAE (raw search-volume)	R ²
ElasticNet	0.568	2.05	0.222
Lasso	0.598	2.01	0.137
Ridge	0.608	2.07	0.108
OLS (usdtry only)	0.608	2.18	0.107
Persistence baseline	0.711	2.71	-0.22

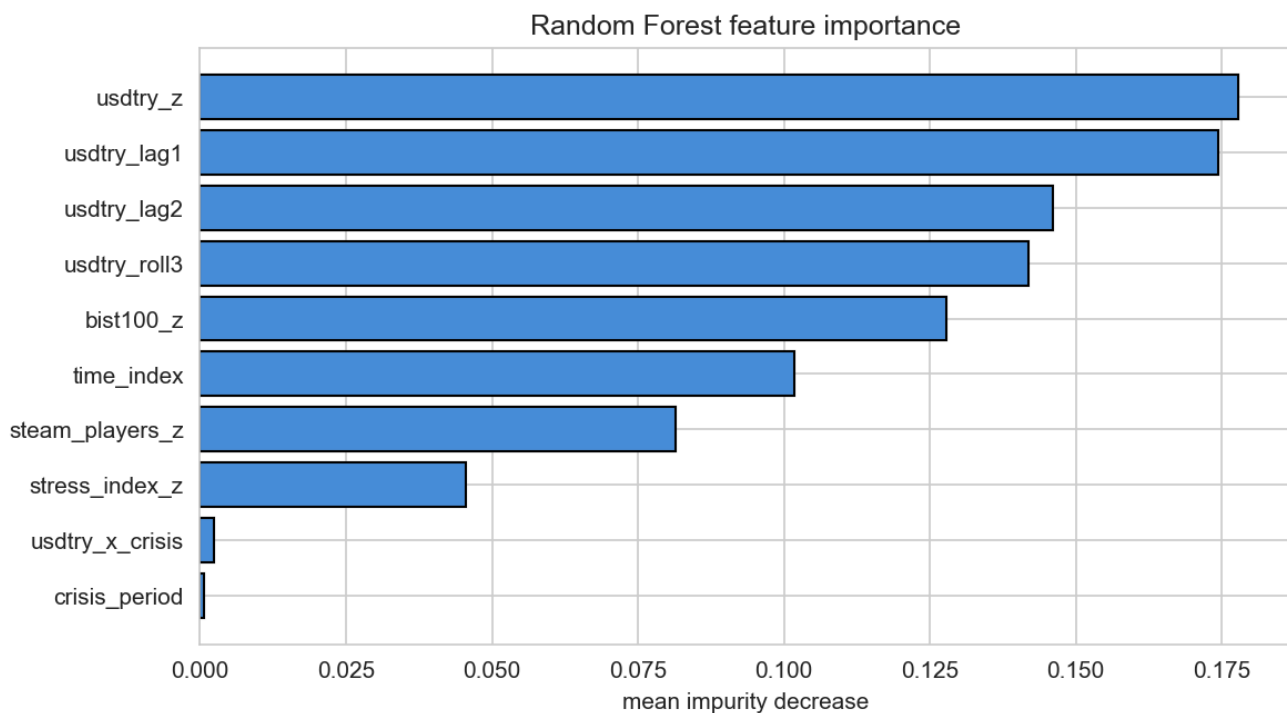
Model	RMSE (z)	MAE (raw search-volume)	R ²
XGBoost	0.736	2.49	-0.31
Random Forest	0.754	2.62	-0.37
Mean baseline	1.220	4.51	-2.59
Linear (no reg.)	1.775	7.44	-6.61

6.4 Predictions vs. actual on the holdout



6.5 What the model relies on





Linear coefficients and permutation importance agree: `usdtry_z` and its two lags carry essentially all the predictive signal; `stress_index_z` is near zero (consistent with H3 being unsupported); `crisis_period` is nearly redundant with the continuous USD/TRY value.

7. Findings

- Macroeconomic depreciation is associated with higher Turkey-specific Steam search interest.** Pearson $r = 0.771$ over 36 monthly observations (H1, $p < 0.001$).
- The effect is also observed as a regime shift**, not just a linear trend: crisis months (USD/TRY $\geq$ 75th percentile) show significantly higher gaming interest than normal months (H2, Mann-Whitney $p < 0.001$).
- Search-volume-based “stress” is a weaker signal than market price.** The composite stress index built from Turkish crisis keywords does *not* correlate with the gaming proxy (H3, $p = 0.76$), and the ML stage gives it ≈ 0 permutation importance.
- A regularised linear model is the best predictor at this sample size.** ElasticNet beats persistence by $\sim 20\%$ on RMSE and is the only non-baseline model with positive R^2 on the chronological holdout. Tree models (Random Forest, XGBoost) underperform on $n \approx 28$ — a textbook small-data result.

Taken together, the EDA and ML stages give **converging support for the digital-escapism hypothesis** in the Turkish 2023–2025 window, with the caveat that the effect is detectable through *behaviour proxies* (Google Trends), not necessarily through people’s *expressed* stress.

8. Limitations

- Proxy target.** `steam_trends_z` is a search-interest proxy for actual Steam playtime in Turkey. Steam does not publish country-level concurrent users, so direct measurement is impossible with

public data.

2. **Small n.** 36 monthly observations is at the low end for ML; tree ensembles in particular cannot be trained reliably.
3. **Single chronological holdout.** A 6-month holdout is one realisation. Different windows would give different point estimates.
4. **Pre-split normalisation.** `_z` columns and `crisis_period` were computed on the full 2023–2025 window during data processing. This keeps the ML stage consistent with the EDA, but a stricter forecasting setup should re-fit z-scores and the crisis threshold using only the training period.
5. **Limited feature space.** No seasonality, no news shocks, no sentiment data, and only short USD/TRY lags. Larger macro models (interest rate, CPI, employment) and calendar effects are not included.
6. **Correlation, not causation.** All findings here are observational. A causal claim (“currency depreciation *causes* more gaming”) would require an identification strategy this project does not attempt.
7. **Stress-index keyword pipeline.** `data_collection.py` requests the keyword `issizlik` (ASCII spelling), while `data_processing.py` looks for the column `işsizlik` (Turkish spelling with `ş`) when building the composite stress index. As shipped, the unemployment keyword is therefore dropped during aggregation and the stress index is built from three of the four planned keywords (`ekonomik kriz`, `enflasyon`, `dolar kur`). The reported H3 result already reflects this — H3 is *not supported* in either case — but a future revision should make the two scripts agree on a single spelling so all four keywords are used.

9. Future Work

- **Daily / weekly granularity** instead of monthly, to better exploit the SteamDB and yfinance data and recover lag dynamics.
- **Country-level Steam telemetry** — partnership / API access to Steam data would replace the search-interest proxy with a direct measure.
- **Multivariate time-series models** (VAR, state-space, structural breaks) suited for joint dynamics of macro + behavioural series.
- **Richer macro features:** TÜFE inflation, central-bank policy rate, unemployment claims, consumer confidence index.
- **Turkish-language sentiment data** from news headlines or social media to replace the brittle “stress keyword” Trends index used here.
- **Causal-style robustness checks:** placebo windows, leave-one-month- out, comparison with a non-crisis country.

10. Reproducibility

```
# 1. Install dependencies
pip install -r requirements.txt

# 2. Collect raw data
python data_collection.py

# 3. Merge, clean, normalise, build the dataset
```

```
python data_processing.py

# 4. EDA + hypothesis tests
jupyter notebook 34314_AliEfe0kudan_EDA.ipynb

# 5. ML stage
jupyter notebook ML/34314_AliEfe0kudan_ML.ipynb
```

Notebooks fix `random_state=42` where applicable, so reported numbers are deterministic.

11. AI Usage

AI assistance (Claude, Anthropic) was used as a coding-support and drafting tool. Disclosure:

- **EDA stage** — see the “AI Usage Log” section inside `34314_AliEfe0kudan_EDA.ipynb`.
- **ML stage** — see `ML/AI_USAGE.md` for a task-by-task breakdown.
- **Final-report stage** — see `Final/AI_USAGE_FINAL.md` (this report’s outline and prose were drafted with AI assistance; every claim, number, table, and figure was verified against the underlying code and data by the author before inclusion).

All research questions, hypotheses, modelling decisions, and conclusions are the author’s. No AI output was kept without verification.

12. Repository Map (final state)

```
DSA210Proj/
├─ data_collection.py
├─ data_processing.py
├─ main.py                                # small utility / scratch
├─ requirements.txt
├─ README.md                             # quick-look entry point
├─ .gitignore
├─
├─ 34314_AliEfe0kudan_EDA.ipynb          # EDA + hypothesis tests
├─ DSA210Proj.pdf                        # proposal PDF
├─
├─ merged_dataset.csv                    # monthly, cleaned, z-scored
├─ usdtry_2023_2025.csv                  # raw USD/TRY
├─ bist100_2023_2025.csv                 # raw BIST100
├─ google_trends_TR.csv                  # raw Trends (TR)
├─ steam_cs2_clean.csv                   # cleaned SteamDB CS2
├─ steamdb_chart_730.csv                 # raw SteamDB CS2
├─ fig_*.png                             # EDA figures
├─
├─ ML/                                  # milestone 3
│   └─ 34314_AliEfe0kudan_ML.ipynb
│   └─ 34314_AliEfe0kudan_ML.html
│   └─ ml_features.csv
│   └─ fig_ml_*.png
```

```

├── AI_USAGE.md
├── ML_Project/                                # beginner-style companion notebook
│   ├── ml_implementation.ipynb
│   ├── build_notebook.py
│   ├── REPORT.md
│   ├── results.csv
│   ├── predictions_vs_actual.png
│   └── feature_importance.png
└── Final/                                    # milestone 4 (this submission)
    ├── FINAL_REPORT.md                      # ← you are here
    ├── FINAL_REPORT.pdf                    # PDF export of this report
    ├── AI_USAGE_FINAL.md                  # AI disclosure for the final stage
    ├── 34314_AliEfeOkudan_ML.html         # static HTML export of the ML notebook
    ├── figures/                           # report-embedded copies of all figures
│   ├── fig_*.png                         # EDA figures
│   └── fig_ml_*.png                      # ML figures
    └── data/                             # snapshot of analysis-ready CSVs
        ├── merged_dataset.csv
        └── ml_features.csv

```

The `Final/` folder is **self-contained**: a grader can read the report and inspect the embedded figures + dataset snapshots without leaving the folder. The rest of the repository is preserved as-is for traceability of each earlier milestone.